**Full Name:** ______________________________

**EEL 3135 – Classwork #04**

**Milestone:**

- Complete any 1 question and 2 parts of a second question to receive credit.

**Question #1:** Consider

$$x(t) = 6 + 10\cos\Big(240\pi t - \frac{\pi}{6}\Big) + 5\cos\Big(360\pi t + \frac{\pi}{4}\Big).$$

(a) Convert each cosine's angular frequency to Hz and list $f_1, f_2$. Determine if $x(t)$ is periodic by checking if $f_1 = K_1 f_0$ and $f_2 = K_2 f_0$ for integers $K_1, K_2$ with $\gcd(K_1, K_2) = 1$. Report $f_0,\ \omega_0,\ T_0$.

(b) Append a third term $4\cos(600\pi t)$. Re-test periodicity and recompute $f_0,\ \omega_0,\ T_0$ if periodic. Comment on how adding terms can break or restore periodicity.

(c) **MATLAB:** Write a script that:
- Defines $x(t)$ over $t \in [0, 4T_0]$ with $f_s = 20\text{kHz}$,
- Plots $x(t)$ and visually verifies periodicity over multiple periods (add grid, labels),
- Overlays vertical lines every $T_0$.

(d) Briefly explain why choosing a smaller valid $f_0$ (when possible) can change a "non-periodic" result to a periodic one.

**Question #2:** Sum/Product & Amplitude Modulation

Let $f_c = 400$ Hz and $\Delta = 20$ Hz:

$$x_{\text{sum}}(t) = \cos\big(2\pi(f_c - \Delta)t\big) + \cos\big(2\pi(f_c + \Delta)t\big).$$

(a) Use complex exponentials to show:

$$x_{\text{sum}}(t) = 2\cos(2\pi\Delta t)\cos(2\pi f_c t).$$

Then state $f_0$ conditions based on $\gcd(K-1, K+1)$ where $f_c = K\Delta$.

(b) **MATLAB:** Plot $x_{\text{sum}}(t)$ for $t \in [0, 0.5]$ s with $f_s = 10$kHz. In separate figures, plot $\cos(2\pi\Delta t)$ and $\cos(2\pi f_c t)$. Label clearly and add legends.

(c) Let $v(t) = 3 + 2.5\cos(60\pi t)$ and $x(t) = v(t)\cos(2\pi \cdot 150\, t)$. Expand $x(t)$ into cosines (carrier + sidebands) and sketch/label the spectrum.

(d) **MATLAB:** Generate $x(t)$ numerically and stem-plot its spectrum. Verify amplitudes match your algebra.

**Question #3:** Orthogonality & Reconstruction

Consider the periodic signal

$$x(t) = 4 + 6\cos\left(\frac{\pi}{4}t\right) + 3\sin\left(\frac{\pi}{3}t\right).$$

(a) Rewrite $x(t)$ using inverse Euler to express $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$, identify $\omega_0$, and list the nonzero $a_k$.

(b) State the Fourier series coefficient definition:

$$a_k = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-jk\omega_0 t}\,dt$$

and explain why orthogonality justifies this extraction.

(c) **MATLAB:** Compute $a_k$ numerically for $|k| \leq 6$ using `trapz` over one period. Display a table of $k$ vs. $a_k$ (real & imag parts). Compare with your analytic values.

(d) **MATLAB:** Make a two-sided `stem` plot of $|a_k|$ for $|k| \leq 6$. Clearly label the nonzero coefficients.

(e) Comment briefly on why coefficients at unused harmonics are $\approx 0$ (numerical orthogonality).

**Question #4:** Fourier Coefficients, Spectrum & Reconstruction

Consider the periodic signal

$$x(t) = 5 + 6\cos\left(\frac{\pi}{5}t - \frac{\pi}{6}\right) - 4\cos\left(\frac{\pi}{3}t + \frac{\pi}{4}\right) + 3\sin\left(\frac{\pi}{2}t\right).$$

(a) Express $x(t)$ as a Fourier Series:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}.$$

Compute all nonzero $a_k$ values by rewriting each term using inverse Euler's formula and combining terms.

(b) MATLAB: Create a two-sided stem plot of $|a_k|$ vs. $k$ for $|k| \leq 6$. Clearly mark $k$ values where $|a_k| = 0$.

(c) Using the $a_k$ coefficients, reconstruct an approximation of $x(t)$:

$$\hat{x}(t) = \sum_{k=-6}^{6} a_k e^{jk\omega_0 t}.$$

Plot both $x(t)$ and $\hat{x}(t)$ on the same graph over $t \in [0, 2T_0]$ and comment on how well they match.

(d) Construct sinusoids from the following Fourier coefficients $a_k$. $a_0 = 2,\ a_{-1} = 3,\ a_1 = 3,\ a_{-4} = -4j,\ a_4 = 4j$ (fundamental frequency: $\omega_0 = 40$)